

J.Mukhesh

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Batch-26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING																		
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026																	
Course Coordinator Name		Dr. Rishabh Mittal																		
Instructor(s) Name		<table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Rapelly Nandini</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M.Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> </table>		Mr. S Naresh Kumar	Ms. B. Swathi	Dr. Sasanko Shekhar Gantayat	Mr. Md Sallauddin	Dr. Mathivanan	Mr. Y Srikanth	Ms. N Shilpa	Dr. Rishabh Mittal (Coordinator)	Dr. R. Prashant Kumar	Mr. Ankushavali MD	Mr. B Viswanath	Ms. Rapelly Nandini	Ms. A. Anitha	Ms. M.Madhuri	Ms. Katherashala Swetha	Ms. Velpula sumalatha	Mr. Bingi Raju
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CourseCode	23CS002PC304	Course Title	AI Assisted Coding																	
Year/Sem	III/II	Regulation	R23																	
Date and Day of Assignment	Week1 - Tuesday	Time(s)	23CSBTB01 To 23CSBTB52																	
Duration	2 Hours	Applicable to Batches	All batches																	
Assignment Number: 1.2(Present assignment number)/24(Total number of assignments)																				
Q.No.	Question	Expected Time to complete																		
1	Lab 1: Environment Setup – <i>GitHub Copilot and VS Code Integration + Understanding AI-assisted Coding Workflow</i> Lab Objectives: To install and configure GitHub Copilot in Visual Studio Code.	Week1 - Monday																		

	• To explore AI-assisted code generation using GitHub Copilot. • To analyze the accuracy and effectiveness of Copilot's code suggestions. • To understand prompt-based programming using comments and code context Lab Outcomes (LOs): After completing this lab, students will be able to: • Set up GitHub Copilot in VS Code successfully. • Use inline comments and context to generate code with Copilot. • Evaluate AI-generated code for correctness and readability. • Compare code suggestions based on different prompts and programming styles.	
	Task 0 • Install and configure GitHub Copilot in VS Code. Take screenshots of each step. Expected Output • Install and configure GitHub Copilot in VS Code. Take screenshots of each step.	
	Task 1: AI-Generated Logic Without Modularization (Factorial without Functions) • Scenario You are building a small command-line utility for a startup intern onboarding task. The program is simple and must be written quickly without modular design. • Task Description Use GitHub Copilot to generate a Python program that computes a mathematical product-based value (factorial-like logic) directly in the main execution flow, without using any user-defined functions. • Constraint: ➤ Do not define any custom function ➤ Logic must be implemented using loops and variables only • Expected Deliverables ➤ A working Python program generated with Copilot assistance ➤ Screenshot(s) showing: ➤ The prompt you typed	

	➤ Copilot's suggestions ➤ Sample input/output screenshots ➤ Brief reflection (5–6 lines): ➤ How helpful was Copilot for a beginner? ➤ Did it follow best practices automatically?	
	Task 2: AI Code Optimization & Cleanup (Improving Efficiency) ❖ Scenario Your team lead asks you to review AI-generated code before committing it to a shared repository. ❖ Task Description Analyze the code generated in Task 1 and use Copilot again to: ➤ Reduce unnecessary variables ➤ Improve loop clarity ➤ Enhance readability and efficiency Hint: Prompt Copilot with phrases like <i>"optimize this code", "simplify logic", or "make it more readable"</i> ❖ Expected Deliverables ➤ Original AI-generated code ➤ Optimized version of the same code ➤ Side-by-side comparison ➤ Written explanation: ▪ What was improved? ▪ Why the new version is better (readability, performance, maintainability).	
	Task 3: Modular Design Using AI Assistance (Factorial with Functions) ❖ Scenario The same logic now needs to be reused in multiple scripts. ❖ Task Description Use GitHub Copilot to generate a modular version of the program by: ➤ Creating a user-defined function ➤ Calling the function from the main block ❖ Constraints ➤ Use meaningful function and variable names ➤ Include inline comments (preferably suggested by Copilot) ❖ Expected Deliverables ➤ AI-assisted function-based program ➤ Screenshots showing: ○ Prompt evolution	

	○ Copilot-generated function logic ➤ Sample inputs/outputs ➤ Short note: ○ How modularity improves reusability.	
	Task 4: Comparative Analysis – Procedural vs Modular AI Code (With vs Without Functions) ❖ Scenario As part of a code review meeting, you are asked to justify design choices. ❖ Task Description Compare the non-function and function-based Copilot-generated programs on the following criteria: ➤ Logic clarity ➤ Reusability ➤ Debugging ease ➤ Suitability for large projects ➤ AI dependency risk ❖ Expected Deliverables Choose one: ➤ A comparison table OR ➤ A short technical report (300–400 words).	
	Task 5: AI-Generated Iterative vs Recursive Thinking ❖ Scenario Your mentor wants to test how well AI understands different computational paradigms. ❖ Task Description Prompt Copilot to generate: An iterative version of the logic A recursive version of the same logic ❖ Constraints Both implementations must produce identical outputs Students must not manually write the code first ❖ Expected Deliverables Two AI-generated implementations Execution flow explanation (in your own words) Comparison covering: ➤ Readability ➤ Stack usage ➤ Performance implications	

	➤ When recursion is <i>not</i> recommended.	
	Submission Requirements 1. Generate code for each task with comments. 2. Screenshots of Copilot suggestions. 3. Comparative analysis reports (Task 4 and Task 5). 4. Sample inputs/outputs demonstrating correctness. Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.	